Jaymin Sheladia

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Education

University of Southern California Los Angeles, USA
Master of Science in Computer Science January 2026 – December 2027
- **Coursework:** Analysis of Algorithms, Agentic AI, Applied NLP, Web Technologies, Database Systems

Graphic Era Deemed to be University Dehradun, India
B.Tech in Computer Science and Engineering October 2021 – June 2025
- **Coursework:** Operating Systems, Network & System Security, Computer Networks, OOPs, Software Engineering

Technical Skills

Languages: C++, Python, Java, C, Go, JavaScript/TypeScript, SQL, HTML/CSS
Systems & Databases: PostgreSQL, MongoDB, Redis , MySQL, Databases Internals, Query Optimization, Linux
Frameworks & Tools: React, Node.js, FastAPI, REST APIs Express.js, Flask, Next.js, Angular, GraphQL, GDB, GCC
Cloud & DevOps: AWS (EC2, S3, Lambda, RDS), Docker, Kubernetes, Firebase, Git, CI/CD, Microservices
AI/ML: PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, RAG, SpaCy, YOLOv11, OpenAI APIs, Claude APIs

Work Experience

Cognizant Technology Solutions May 2025 – September 2025
Programmer Analyst Trainee – Software Engineering Coimbatore, India
- Engineered a full-stack Travel Planner and Booking System using Spring Boot and Angular, designing and integrating **10+** RESTful APIs to centralize itinerary, booking, and user preference data across the booking workflow.
- Integrated a Generative AI-based recommendation engine that evaluates **3** key signals – user interests, budget, and trip duration – to generate context-aware personalized travel recommendations for users.

CODTECH IT SOLUTIONS June 2024 – August 2024
Machine Learning Intern Remote
- Designed and trained a fraud detection model using anomaly detection and supervised machine learning on over **200,000** credit card transactions, reaching **98%** classification accuracy on live transaction data.
- Optimized machine learning algorithms and re-engineered data preprocessing pipelines for production models, improving predictive accuracy by **12–15%** across multiple live projects and reporting dashboards.

Projects

Model Router | GitHub | Python, FastAPI, SQLite July 2026 – August 2026
- Architected a policy-aware LLM routing service in FastAPI, layering identity verification, per-project authorization, budget enforcement, and automatic model failover, with every decision logged to a SQLite audit trail.
- Benchmarked the routing classifier against a hand-labeled oracle set using live model inference, reaching **100%** classification accuracy while cutting inference cost by **27.9%** versus always calling the top-tier model.

AI Research Assistant | GitHub | Python, FastAPI, Next.js, RAG June 2026 – July 2026
- Engineered a full-stack AI research assistant using FastAPI, Next.js, and PostgreSQL with pgvector, implementing a RAG pipeline for semantic retrieval, contextual Q&A, and citation knowledge-graph traversal across ingested papers.
- Profiled and benchmarked the end-to-end document ingestion pipeline across live LLM inference, embedding, and persistence stages, achieving latency below **4.3 seconds** per paper, with LLM inference as **85–95%** of total latency.

Sequence Alignment: DP vs. Divide & Conquer | GitHub | Python March 2026 – April 2026
- Implemented and comparatively evaluated classical Needleman-Wunsch dynamic programming against Hirschberg's divide-and-conquer algorithm for global DNA sequence alignment, preserving optimal alignment correctness.
- Validated the theoretical $\mathbf{O(m \times n) \rightarrow O(m+n)}$ memory reduction empirically, confirming up to **90%** lower peak memory usage with bit-for-bit identical optimal alignments across increasing input sizes.

Research

Deep Learning Approach for Malicious URL Detection | Co-author; Published Research Paper 2025
- Compared **4** deep learning architectures (CNN, RNN, LSTM, Bi-LSTM) for malicious URL detection, engineering character-level preprocessing pipelines on the IEEE Dataport dataset for sequential analysis at scale.
- Achieved **99.25%** accuracy with the Bi-LSTM model, outperforming CNN (**86.5%**), LSTM (**63.75%**), and RNN (**53%**) baselines across standard classification and error-rate metrics.